



Technical Note

Self-labelled encoder-decoder (SLED) for multi-echo gradient echo-based myelin water imaging

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ABSTRACT

Purpose: Reconstruction of high quality myelin water imaging (MWI) maps is challenging, particularly for data acquired using multi-echo gradient echo (mGRE) sequences. A non-linear least squares fitting (NLLS) approach has often been applied for MWI. However, this approach may produce maps with limited detail and, in some cases, sub-optimal signal to noise ratio (SNR), due to the nature of the voxel-wise fitting. In this study, we developed a novel, unsupervised learning method called self-labelled encoder-decoder (SLED) to improve gradient echo-based MWI data fitting.

Methods: Ultra-high resolution, MWI data was collected from five mouse brains with variable levels of myelination, using a mGRE sequence. Imaging data was acquired using a 7T preclinical MRI system. A self-labelled, encoder-decoder network was implemented in TensorFlow for calculation of myelin water fraction (MWF) based on the mGRE signal decay. A simulated MWI phantom was also created to evaluate the performance of MWF estimation.

Results: Compared to NLLS, SLED demonstrated improved MWF estimation, in terms of both stability and accuracy in phantom tests. In addition, SLED produced less noisy MWF maps from high resolution MR microscopy images of mouse brain tissue. It specifically resulted in lower noise amplification for all mouse genotypes that were imaged and yielded mean MWF values in white matter ROIs that were highly correlated with those derived from standard NLLS fitting. Lastly, SLED also exhibited higher tolerance to low SNR data.

Conclusion: Due to its unsupervised and self-labeling nature, SLED offers a unique alternative to analyze gradient echo-based MWI data, providing accurate and stable MWF estimations.

1. Introduction

Myelin water imaging (MWI) (Whittall and MacKay, 1989; Mackay et al., 1994) is an established quantitative magnetic resonance imaging (MRI) technique that has been extensively applied to study such myelin related diseases in human brain and spinal cord as multiple sclerosis (Vargas et al., 2015; Laule et al., 2010), neuromyelitis optica spectrum disorder (Manogaran et al., 2016), progressive solitary sclerosis (Lee et al., 2019), Parkinson's disease (Baumeister et al., 2019), stroke (Borich et al., 2013), schizophrenia (Lang et al., 2014), and cervical spondylotic myelopathy (Liu et al., 2017). Traditionally, MWI is

applied by collecting multi-echo spin echo signals that can be evaluated to extract the fast-decaying MR signal of myelin water (water trapped between myelin bilayers). The myelin water decay signal is typically computed from the total MR signal based on its characteristic short T_2 relaxation time. Specifically, the ratio of myelin water signal to the total signal is termed myelin water fraction (MWF). It has been extensively applied as an imaging biomarker to quantify myelin in the human central nervous system (Laule et al., 2006, 2008).

Unlike conventional T_2 MWI where refocusing radiofrequency (RF) pulses are applied to acquire spin echoes, multi-echo gradient echo based MWI (Du et al., 2007) samples the multi-component T_2^* de-

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cay using unipolar or bipolar gradient echoes. This method has become increasingly popular for its high data acquisition efficiency and low specific absorption rate (SAR) (Lee et al., 2021). However, it can result in somewhat lower relative signal-to-noise ratio (SNR) and be sensitive to B_0 inhomogeneity. T_2^* MWI is also affected by the white matter fiber orientation, particularly for fibres perpendicular to B_0 (Rudko et al., 2014). Nevertheless, since it does not employ large flip angle RF refocusing RF pulses, T_2^* MWI generally avoids flip angle errors due to B_1^+ inhomogeneity, stimulated echoes, SAR limitations, and longer scan times. These technical challenges are present in T_2 MWI techniques, especially at high magnetic field strengths. We believe multi-echo gradient echo (mGRE) T_2^* MWI has the potential to be particularly useful at high fields where SNR is increased, making it a suitable choice for our current 7 T pre-clinical MRI study.

Conventionally, T_2^* decay data acquired by a mGRE sequence has been fit using a nonlinear least squares (NLLS) algorithm with a simplified three-pool model. Such a model typically accounts for signal contributions from myelin water, axonal water, and other water pools (Du et al., 2007). Depending on whether magnitude or complex MRI data are used, the three-pool NLLS method is usually designed to fit either six free parameters (T_2^* relaxation time and signal amplitude for each pool) or ten parameters (T_2^* time, frequency shift, and amplitude for each pool, and an initial phase offset) (Hwang et al., 2010; Nam et al., 2015). However, fitting the NLLS model to data in this way is very susceptible to noise, dependent on the pre-determined T_2^* range of each pool, and a function of the manually selected initialization parameters. Consequently, it can lead to unstable and inaccurate fitting results. For this reason, alternatives to the conventional NLLS fitting strategy for mGRE myelin water imaging are highly desirable.

In recent years, supervised machine learning (ML) and deep learning (DL) methods have been successfully implemented as important tools for many MR applications (Lundervold and Lundervold, 2019). Frequently, supervised ML or DL approaches for quantitative MRI parameter estimation involve three steps: (1) application of a specific signal model to generate a comprehensive library of simulated MR signals, (2) training of a ML or DL model to learn the non-linear mapping between the simulated signals and ground truth parameters, (3) deployment of the trained model to experimental data to obtain quantitative parameter maps. For MWI, implementation of these modern methods has largely been focused on analyzing multi-exponential T_2 decay signals collected using multi-echo spin echo sequences (Lee et al., 2020; Liu et al., 2020, 2022; Nagtegaal et al., 2020; Kumar et al., 2018; Yu et al., 2021; Li et al., 2021). Despite similarities between the T_2 and T_2^* decay characteristics, only a limited number of studies have developed ML or DL based analysis strategies for measuring multi-exponential T_2^* decay. Recently, Jung et al. (2021) introduced an artificial neural network (ANN) (Schmidhuber, 2015) with superior MWF estimation accuracy compared to conventional NLLS. Later, these authors extended their findings using a complex-valued ANN to analyze the complex T_2^* decay signal obtained from human brain 3 T MRI scans (Jung et al., 2022).

Despite their theoretical advantages in model generalization and overfitting prevention compared to their supervised counterparts, unsupervised ML and DL methods for MWF estimation have yet to be robustly applied (Rahbek et al., 2021). As a result, in this study, we propose an unsupervised approach, called self-labelled encoder-decoder (SLED), to calculate the MWF from multi-echo gradient echo-derived T_2^* decay. SLED first utilizes a series of neural networks to encode the multi-echo magnitude input data in a latent space where MWF can be extracted. Next, a three-pool signal model is applied to decode the latent space back to the multi-echo image space, where the input data are regarded as the ground truth labels for model training. Since the input data are applied as ground truth labels, the SLED method is self-labelled and unsupervised. SLED is designed to discover the inherent structure of

Table 1

Ex-vivo transgenic, mouse brain samples employed for testing the MWF fitting accuracy of SLED. The enhancer knockout (KO) mice employed in our study were derived from a novel graded hypomyelination mouse panel for which one or more of the M1E, M3, and M5 transcription enhancer domains were deleted from the myelin basic protein/Golli (*mpb/Golli*) locus. In addition to the KO mice, a wild type and a hypermyelinating (M3KO;389bpKI) mouse were also analyzed to investigate the impact of elevated myelin content on MWF fitting accuracy of SLED.

Mouse Line	<i>mpb</i> mRNA (% $\pm$ error)	Abbreviation
M3KO;389bpKI (hypermyelinating)	160 $\pm$ 5	HyperM
Wild type	100 $\pm$ 6	WT
M5KO	63 $\pm$ 4	1KO
M3M5KO	28 $\pm$ 1	2KO
M1EM3M5KO	15 $\pm$ 1	3KO

each input dataset in an unsupervised manner, providing an alternative, unique way to calculate MWF.

2. Methods

2.1. MRI experiment and mouse model

Multi-component, T_2^* decay data was collected using a 24-echo, 3D multi-echo gradient recalled echo (mGRE) sequence with TE/ Δ TE/TR = 2/2/100 ms, bipolar readout, 18 signal averages, matrix size = 135 $\times$ 106 $\times$ 120 and acquired spatial resolution = 100 $\times$ 100 $\times$ 100 μ m³. The even and odd echo k-space misalignment due to the bipolar gradient acquisition was corrected utilizing the methods proposed by Lu et al. (2008) Perfused and fixed (Weil et al., 2019) brains of five young mice, immersed in an MRI-invisible matching liquid (FC-40 Fluorinert), were imaged using a Bruker Pharmascan 7 T MRI system. The mice imaged as part of this study were derived from a novel graded hypomyelination mouse panel for which one or more of the M1E, M3, and M5 transcription enhancer domains were deleted from the myelin basic protein/Golli (*mpb/Golli*) locus (Bagheri et al., 2020). The well-characterized enhancer knockout (KO) mouse lines (Table 1) were specifically employed to benchmark myelin quantification using SLED and NLLS fitting approaches. In addition, a wild type (WT) mouse and a hypermyelinating (HyperM) mouse line, M3KO;389bpKI, were also analyzed (Bagheri et al., 2020). All mouse experiments were reviewed and approved by the Animal Care Committee of the Montreal Neurological Institute, McGill University.

2.2. Self-labelled encoder-decoder (SLED)

The schematic diagram in Fig. 1 depicts the SLED network architecture, constructed using an open-source software library, TensorFlow (Abadi et al., 2016). The SLED encoder was designed to use mGRE data (in our case, 24-echo decay signals) as input data and produce six latent parameters (T_2^* relaxation times and amplitudes of the myelin, axonal, and free water pools) at the encoder output. The encoder consisted of three neural networks with an identical structure (layers: 24 $\times$ 256 $\times$ 128 $\times$ 1, activation: sigmoid (Narayan, 1997)). Each of these networks was used to calculate the three T_2^* relaxation times. Constraints were applied at the output layer of each network to scale the output T_2^* values within specific T_2^* ranges. In particular, the fitting constraints were strategically determined based on a previously published, data-driven approach (Supplementary Materials: 1. Justification for T_2^* constraints) (Liu et al., 2022). One neural network (layers: 24 $\times$ 256 $\times$ 128 $\times$ 3, activation: sigmoid for the hidden layers and softmax (Maida, 2016) for the output layer) was applied to calculate the amplitudes of the three pools. The justification for each neural network's configuration was provided in the Supplementary Materials: 2. SLED configuration experiment.

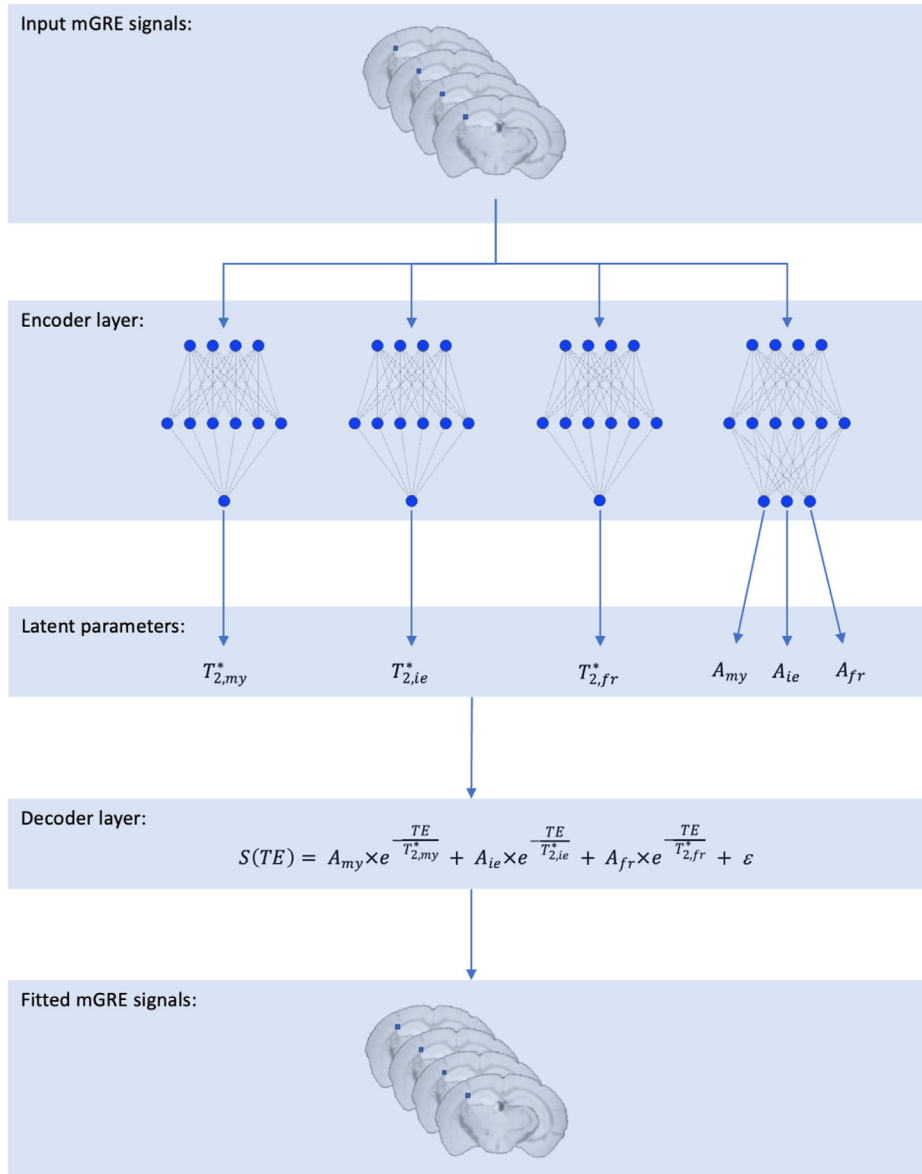


Fig. 1. Schematic diagram of self-labelled encoder-decoder (SLED). The encoder step of SLED utilizes several neural networks to encode the input multi-echo gradient-echo (mGRE) data and output six latent parameters. The latent parameters represent the T_2^* relaxation times and the amplitudes of each water pool. The decoder step of the network uses a three-pool model to decode the latent parameters and fit them to the original, experimental mGRE data.

The physics-modelled decoder for SLED was initiated with the six latent parameters obtained from the encoder. It was designed to yield 24-echo decay signals using the following three-pool exponential decay model:

$$S(TE) = A_{my} e^{-\frac{TE}{T_{2,my}^*}} + A_{ie} e^{-\frac{TE}{T_{2,ie}^*}} + A_{fr} e^{-\frac{TE}{T_{2,fr}^*}} \quad (1)$$

where A_{my} , A_{ie} , A_{fr} and $T_{2,my}^*$, $T_{2,ie}^*$, $T_{2,fr}^*$ denote the corresponding amplitude and T_2^* times of the myelin water, intra and extracellular (IE) water, and free water pools. Subsequently, $S(TE)$ was mixed with orthogonal, zero mean Gaussian noise (Cárdenas-Blanco et al., 2008) to simulate varying noise levels in the magnitude MR signal $S(TE)_{noisy}$:

$$S(TE)_{noisy} = \sqrt{(S(TE) + \text{Gaussian noise})^2 + (\text{Gaussian noise})^2} \quad (2)$$

For model testing, the noise level was adjusted to yield mGRE decay data with different SNR levels uniformly sampled from a SNR range from 50 to 500 for each signal.

SLED was then applied to the ultra-high resolution (100-micron isotropic voxels) *ex-vivo* mGRE data collected for each mouse brain. The SLED algorithm was self-trained using the decay signals of brain-only voxels as input data to the encoder and ground truth labels of the fitted signals from the decoder. The loss function for the SLED network was

specified based on the mean squared error (MSE) between the fitted signals extracted from the decoder (Eq. (1)) and the experimental mGRE decay data. The training objective was to minimize the losses of all training examples for each mouse brain. The numerical minimization step was carried out using the Adam optimizer (Kingma and Adam, 2017). The optimization updated six latent parameters (A_{my} , A_{ie} , A_{fr} , $T_{2,my}^*$, $T_{2,ie}^*$, and $T_{2,fr}^*$) at each iteration. The iterative training process was stopped when the training loss no longer decreased. Once trained, the MWF value was calculated based on the corresponding latent parameters for each brain image voxel (Eq. (2)).

$$MWF = \frac{A_{my}}{A_{my} + A_{ie} + A_{fr}} \quad (3)$$

Note that SLED was trained exclusively for each subject, in contrast to supervised learning approaches such as the ANN where the trained model is applied universally to all subjects with proper data normalization.

2.3. Nonlinear least squares (NLS)

We compared the SLED model to a traditional NLS fitting approach. Application of three-pool NLS fitting to extract MWF from mGRE mag-

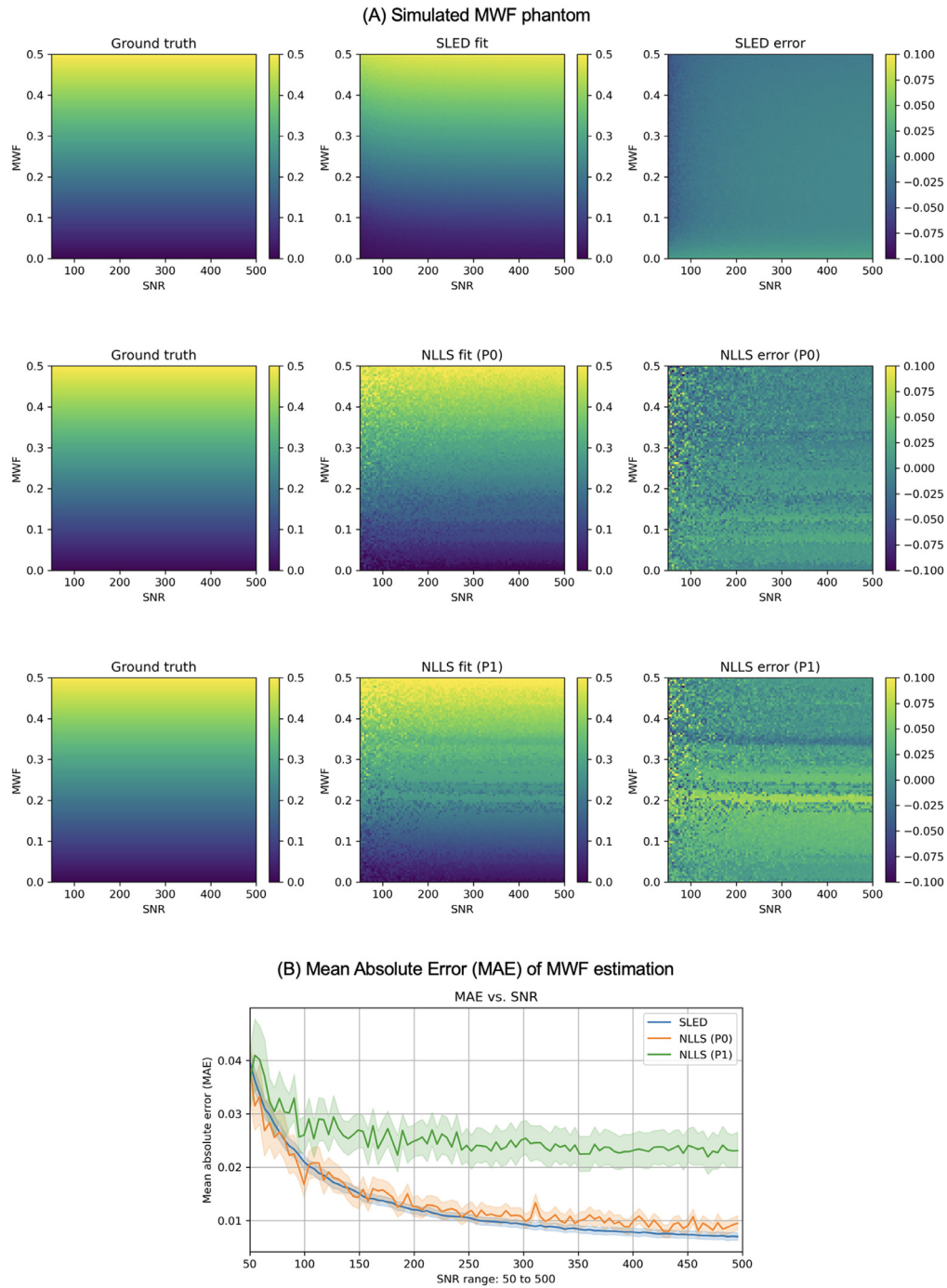


Fig. 2. *In silico* MWF phantom results. (A) A 2D MWF phantom (100×100 voxels) with gradually increasing SNR levels (horizontal axis) and MWF values (vertical axis.) The *in silico* phantom was created to test MWF estimation accuracy for the SLED and NLLS algorithms (first row: SLED, second row: NLLS (P0), third row: NLLS (P1)). (B) Mean absolute error (MAE) of fitted MWF (with 95% confidence intervals) is plotted against mGRE image SNR.

nitude data has previously been illustrated (Hwang et al., 2010). The fitting parameters of our specific, in-house NLLS implementation were adaptively selected to match the requirements of each of our experiments.

2.4. Simulated phantom tests

A two-dimensional (2D), *in-silico*, MWF phantom (consisting of 100×100 voxels) was created to test the fitting accuracy of both SLED and NLLS. The phantom was initialized with SNR levels ranging from

50 to 500 in step sizes of 4.5 (shown from left to right in Fig. 2). The vertical axis of the phantom from bottom to top corresponds to MWF values gradually increasing from 0 to 0.5. Other selected parameters for generating the simulated phantom, such as the T_2^* time and the amplitude of each water pool, are summarized in Table 2. For each voxel in the phantom, 24 gradient echo decay signals were generated according to Eq. (1). The simulated phantom decay curves were then analyzed using SLED and NLLS, respectively, to calculate MWF maps. NLLS fitting parameters are summarized in Table 2. Mean absolute error (MAE) with a 95% confidence level was used to evaluate the MWF fitting ac-

Table 2

Parameter list demonstrating the spatial dimensions and parameter initializations chosen for the *in silico*, simulated MWF phantom. The phantom was used for fitting the SLED and NLLS models. Only NLLS required initialization of parameters for fitting.

Simulated MWF phantom parameters	
Number of voxels in X dimension	100
Number of voxels in Y dimension	100
SNR range in X dimension	[50, 500]
MWF range in Y dimension	[0, 0.5]
Free water fraction (FWF)	0.05
Intra and extracellular (IE) water fraction	1 – MWF – FWF
T2s of myelin, IE, and free water pools	[0.01, 0.05, 0.25] seconds
SLED and NLLS fitting constraints	
Amplitude range for the 3 pools	[0, 1]
T2 range for myelin water pool	[0.003, 0.015] seconds
T2 range for IE water pool	[0.045, 0.07] seconds
T2 range for free water pool	[0.2, 0.3] seconds
Initialization parameters for NLLS (P0 set)	
Amplitudes of myelin, IE, and free water pools	[0.25, 0.7, 0.05]
T2s of myelin, IE, and free water pools	[0.01, 0.05, 0.25] seconds
Initialization parameters for NLLS (P1 set)	
Amplitudes of myelin, IE, and free water pools	[0.3, 0.6, 0.01]
T2s of myelin, IE, and free water pools	[0.008, 0.055, 0.2] seconds

curacy for each independent SNR level employed as part of the *in-silico* phantom.

To investigate the sensitivity of SLED and NLLS fitting to different initialization parameters, two initialization parameter sets, P0 and P1, were tested. The P0 initialization parameter set was considered ‘optimal’ because it contained the true gold standard parameters used for creating the *in-silico* phantom. P1 was initialized with small deviations from the ‘optimal’ initialization parameter set (Table 2).

2.5. Mouse data analysis

mGRE magnitude data collected based on ultra-high resolution, *ex-vivo* imaging of the mouse brains was first masked using the FSL brain extraction tool (BET) (Zhang et al., 2001). Then, mouse brain MWF maps were calculated using both SLED and NLLS with fitting parameters reported and justified in Supplementary Materials: 1. Justification for T_2^* constraints. The data processing time was recorded (CPU: Intel, i7-11700F @ 2.50GHz, 8 cores). For further statistical analysis, the resulting MWF maps were non-linearly registered into a standard space, the Allen mouse brain atlas space (Wang et al., 2020). Deformation fields, applied for the registration (rigid, affine, and deformable transformation), were obtained by registering the mean image from all gradient echoes to the Allen mouse brain atlas space. The registration was carried out using the advanced normalization tools (ANTs) (Avants et al., 2008). Mean and variance of MWF were evaluated in specific mouse brain regions of interest (ROI), including internal capsule, arbor vitae, cerebellar commissure, anterior commissure (olfactory limb and temporal limb), and corpus callosum (body, genu and splenium). Pearson correlation and paired t-tests were applied to compare SLED and NLLS-derived MWF values for each distinct white matter ROI and for each mouse genotype.

To further investigate the influence of noise on model fitting, particularly in low SNR scenarios, MR noise (Gaussian noise on both real and imaginary channels) was added to the multi-echo gradient echo decay data. Specifically, noise was added to produce different SNR levels (SNRs = 50, 40, 30, 20, 10, 5) for the mGRE data of the wild type (WT) mouse. SLED and NLLS were then applied to calculate MWF at each SNR level. Whole brain mean and standard deviation in MWF MAE at each SNR level were then evaluated as performance measures. A paired t-test was performed to determine whether there was a significant difference ($p < 0.05$) in MAE between SLED and NLLS fitting methods.

2.6. Code availability

To support reproducible research, the code associated with the SLED method is made freely available at the following URL - <https://github.com/hanwencat/SLED>.

3. Results

3.1. Simulated phantom tests

As shown in Fig. 2(A), SLED produced MWF maps that closely matched the ground truth applied for *in-silico* phantom tests. Although SLED MWF values were slightly overestimated for very low (true MWF = 0) and underestimated for very high (true MWF = 0.5) MWF, a smooth transition in calculated MWF values was observed (Fig. 2) from the lowest to the highest values. SLED calculated MWF was devoid of sudden jumps in MWF at all SNR levels. In contrast, NLLS fitting with both P0 and P1 initializations yielded noisier MWF maps. In some cases, application of NLLS fitting resulted in unwanted streaking lines across the entire SNR range of the *in-silico* phantom. This pattern was particularly observed when fitting with NLLS using the P1 initialization.

The MAE of MWF estimation at each SNR level is presented in Fig. 2(B). Higher SNR values are expected to result in increased fitting accuracy, regardless of chosen MWF estimation approach. However, this expected fitting accuracy improvement reached a plateau when the SNR level was 250. As expected, NLLS (P0) performed much better than NLLS (P1), since P0 was the ‘optimal’ fitting initialization. Overall, as compared to NLLS, SLED demonstrated lower MAE at a range of realistic SNR levels (>100). SLED also had the smallest variability in fitting accuracy by a wide margin (as demonstrated by the 95% confidence levels in Fig. 2(B)).

3.2. Mouse brain data

The average time for processing each brain data set was 151.5 ± 2.1 s for SLED and 289.1 ± 5.3 s for NLLS. From a visual inspection, NLLS-derived MWF maps were more influenced by noise, compared to corresponding maps derived from SLED (Fig. 3). The five mouse genotypes we examined included: hyper myelinated (HyperM), WT, M5KO, M3M5KO, and M1EM3M5KO genotypes. We observed a decrease in MWF from the most myelinated (HyperM) to the least myelinated (M1M3M5KO) genotypes using both SLED and NLLS MWF analysis methods (Fig. 4). This was true in almost all the major white matter ROIs examined in Fig. 4. Compared to SLED, NLLS produced higher MWF in all ROIs ($p < 0.05$). Higher MWF values calculated using NLLS may result from the higher sensitivity of NLLS fitting to noise in the magnitude signal decay (Fig. 3). The mean MWF in all ROIs calculated by NLLS and SLED were significantly correlated ($p < 0.05$) and demonstrated a similar pattern across different genotypes. In addition, the WT mouse brain was used to test the performance of both methods for estimating the T_2^* relaxation times of all three water pools, as well as the free water and IE water fractions. The results of these tests are shown in Supplementary Materials: 3. All metric maps.

3.3. Model susceptibility to noise

Fig. 5 identifies the influence of adding simulated, zero-mean Gaussian noise to the experimental mGRE data. In particular, Fig. 5 shows the impact of different levels of input noise on NLLS and SLED MWF reconstructions for a representative coronal slice of a WT mouse brain. Compared to NLLS, SLED demonstrated significantly lower MWF MAEs ($p < 0.05$). This was true for all SNR levels we investigated. Including additional noise in mGRE signal decay did not noticeably affect the SLED MWF calculation (MAEs $\leq 0.0134 \pm 0.0306$) for all the SNR levels with SNR > 10. However, the performance of SLED was reduced (MAEs $\leq 0.0964 \pm 0.1055$) for low SNR levels (SNR ≤ 10). In contrast, NLLS

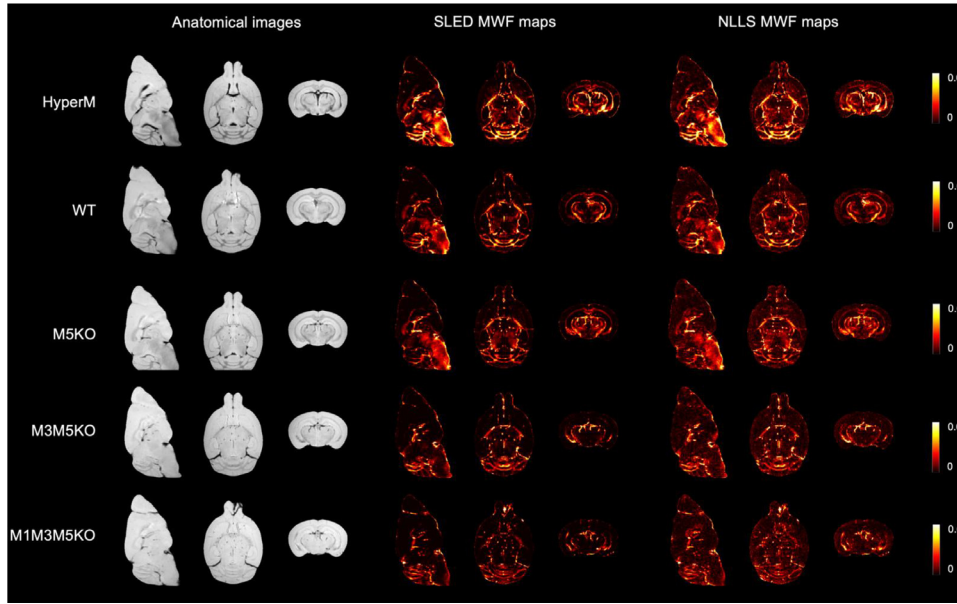


Fig. 3. Mouse brain MWF maps calculated for each mouse genotype imaged as part of this study. Representative three-plane (sagittal, axial and coronal) slices of mouse brain MWF maps are shown for all genotypes, including hyper myelinated (HyperM), wild type (WT), M5KO (1KO), M3M5KO (2KO), and M1EM3M5KO (3KO). Anatomical, three-plane images, averaged across all 24 echoes from the mGRE data sets are shown on the left side of the figure.

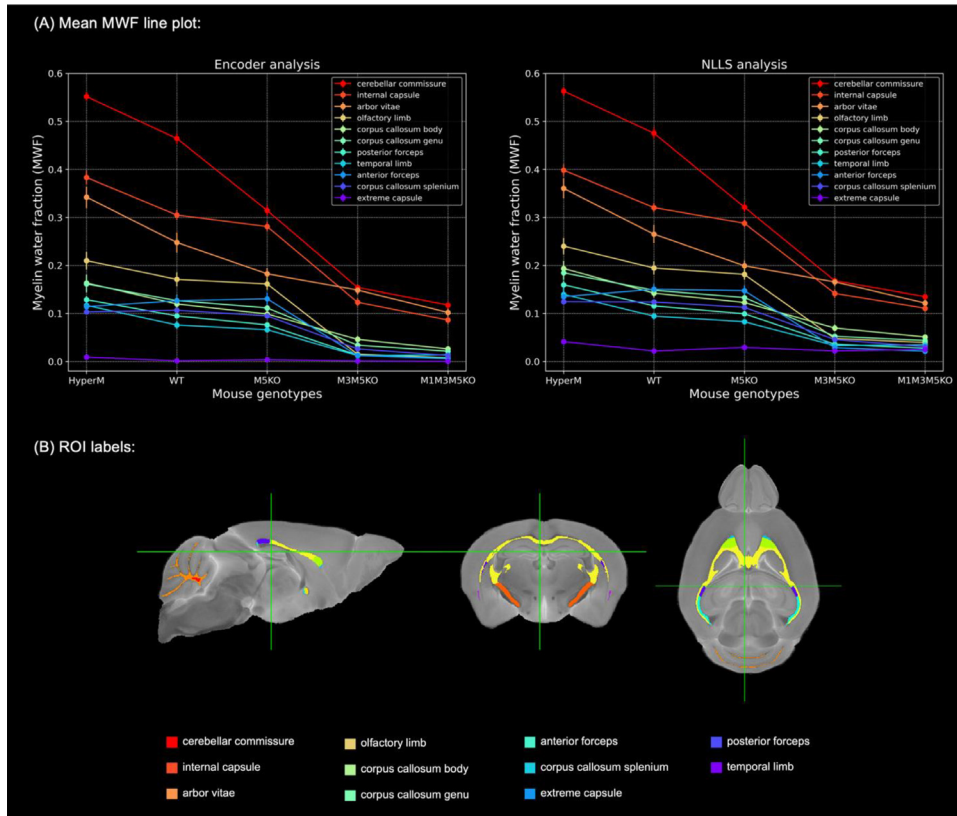


Fig. 4. MWF calculated in white matter ROIs for each of the mouse genotypes scanned as part of this study. (A) The line plots show mean and standard deviation in each white matter tract ROI for hypermyelinated (HyperM), wild type (WT), M5KO (1KO), M3M5KO (2KO), and M1EM3M5KO (3KO) mice. (B) ROI labels are overlaid on the high-resolution, Allen Brain Mouse Atlas template space employed for this study.

MWF calculation showed a gradual degradation of performance as SNR levels decreased from 50 ($MAE = 0.1234 \pm 0.1590$) to 5 ($MAE = 0.1611 \pm 0.1609$).

4. Discussion

4.1. Comparison of SLED and NLLS methods for computing myelin water fraction

SLED and NLLS both seek to minimize a Mean Squared-Error (MSE) loss function which defines the difference between the experimental and

model-fitted signals. However, there are two key differences between the two methods: (1) SLED leverages neural networks to obtain latent parameters such as the T_2^* relaxation times and amplitudes of each pool. It does so with an automatic and randomized initialization (Yam and Chow, 2000). This step avoids handpicking initialization parameters (a feature which is inherent in most NLLS implementations); (2) SLED simultaneously uses the data from all imaging voxels to train the neural networks, while NLLS operates separately for each voxel.

Since SLED leverages neural networks with automatic, randomized initialization, it is more likely to converge to the global minimum of the loss function compared to NLLS. Our phantom tests (Fig. 2) demon-

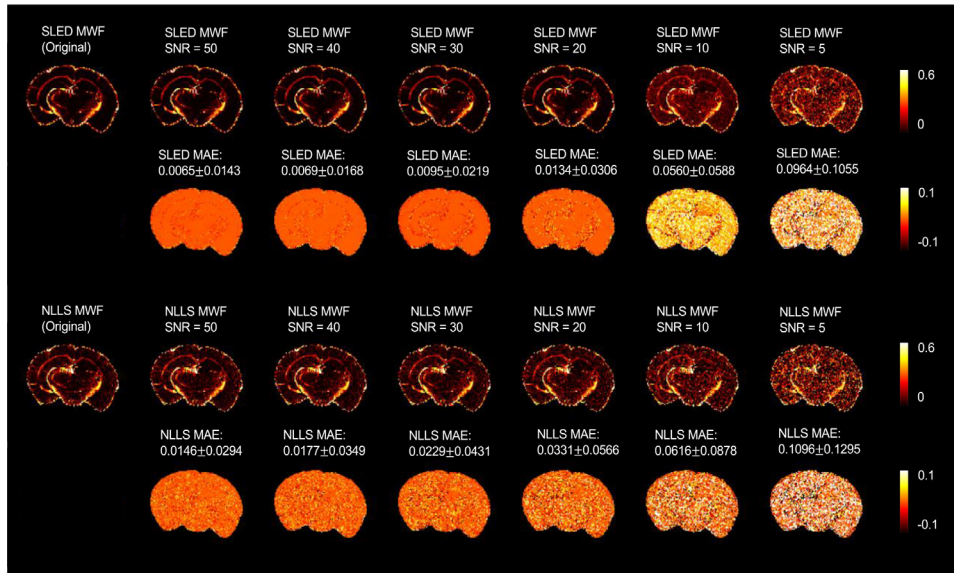


Fig. 5. Effect of noise on MWF fitting accuracy for SLED compared to NLLS. The robustness of SLED and NLLS to noise contamination was tested by injecting simulated, zero-mean Gaussian noise with different SNR levels (SNRs = 50, 40, 30, 20, 10, 5) into the original mGRE data. The original and noise contaminated MWF maps produced by SLED and NLLS are shown. The mean absolute error (MAE) of MWF and corresponding standard deviations are also presented. The MWF outer rim depicts the mouse brain was caused by susceptibility artifacts at the boundary between the edge of the mouse brain and the fluorinated solution in which the fixed mouse brains were immersed for MR scanning.

strated that even small deviations from the ‘optimal’ initialization degrades the performance of NLLS. Moreover, application of the NLLS algorithm can result in identification of local minimums when minimizing a loss function. Although the NLLS MWF mapping shows approximately the same level of accuracy as SLED with the ‘optimal’ initialization, such an ‘optimal’ initialization parameter set cannot always be determined *a priori* when conducting *in-vivo* human or pre-clinical imaging. This is particularly true, for example, in the case of brain pathology. Consequently, we believe that avoiding manual selection of initialization parameters is a major advantage of SLED.

Since SLED simultaneously uses data from all imaging voxels to train its neural networks, it inherently suppresses background noise in MRI data. This can result in MWF maps with a more homogeneous appearance and improved white matter tract detail compared to NLLS fitting (Fig. 3). Additionally, because whole brain mGRE decay data are utilized as inputs for SLED model training, SLED simultaneously learns the non-linear relationship between decay signals and latent parameters from all brain voxels. This is analogous to a form of global regularization (Girosi et al., 1995). In contrast, NLLS fitting with the trust-region-reflective algorithm operates in a voxel-wise manner, making it sensitive to stochastic noise. For voxels with magnitude signal decay impacted by low SNR, NLLS fitting is less accurate. As is well-recognized, low SNR may be encountered, in practical imaging cases, when applying fast imaging (Dvorak et al., 2020) or low-field portable MRI scanners. (Wald et al., 2020; Liu et al., 2021) It is increasingly important to develop data fitting algorithms to accommodate low SNR data. SLED is suited to MWF in more SNR-starved scenarios. Fig. 5 demonstrates that SLED has extremely high tolerance to fitting errors introduced by low SNR levels. This finding is further supported by our additional *in-silico* MWF phantom experiment presented in Supplementary Materials: 5. Simulated MWF phantom for low SNR levels.

4.2. Evaluation of parametric vs. non-parametric methods

The greatest imprecision in applying a parametric method such as NLLS to a conventional 3-pool model is that the fitting results may be biased based on predefined model parameters. Prior information can be very helpful for improving the fitting stability, but it can come at a cost of reduced generality. On the other hand, non-parametric methods, such as the non-negative least squares (NNLS), do not require prior information, making their fitting results more objective. However, non-parametric methods such as NNLS are often mathematically

ill-posed and may have poor stability, even after integrating regularization techniques to smooth the T_2 or T_2^* distributions (Istratov and Vyvenko, 1999). Recently, substantial progress has been made with spin echo-based T_2 MWI to overcome this challenge. Notably, Canales-Rodríguez et al. (2021) provided a comprehensive guide for selecting regularization functions and their weighting factors when applying NNLS to MWI data, and Liu et al. (2022) also proposed a machine learning approach, SAME-ECOS, to improve T_2 MWF fitting accuracy and stability. In our present study, although the 3-pool model adopted in SLED was carefully justified using the resolution limit formalism (Supplementary Materials: 1. Justification for T_2^* constraints), manual selection of some model parameters (such as the T_2^* boundary values for each pool) was necessary. Developing SLED as a truly non-parametric method for the gradient echo based T_2^* MWI is a potential future direction of research.

4.3. Evaluation of supervised vs. unsupervised methods

Because it is unsupervised and self-labelled, SLED requires exclusive self-labeling for each dataset. Although it does not provide a trained model that can be applied universally for all datasets, it circumvents challenges related to model generalization and overfitting (Hawkins, 2004) commonly observed in most ML and DL models. This is a notable advantage of SLED. Unlike supervised methods such as ANNs, a numerically simulated library containing many synthetic MR signals is not required for model training, validation, and testing of SLED. This feature is important because the generation of simulated MR decay data can be time consuming. Moreover, generation of decay signals typically relies on a model that may not be representative of all possible MR signal evolutions or may not account for various imperfections during the signal formation and data collection processes (e.g. influence of B_0 field inhomogeneity and eddy current effects). SLED takes advantage of self-labeling to avoid the aforementioned challenges and preserve the full information contained in each dataset. In addition, SLED offers more flexibility for analysis of other quantitative MRI contrasts. This could be achieved by substituting the current three-pool model with a different signal model, as needed by the decoder. Such a step alleviates the need for re-generating the signal library and re-training an ANN model, as is the case for supervised ANN approaches. For demonstration purposes, a comparison between the unsupervised SLED approach and two supervised artificial neural network methods is presented in Supplementary Materials: 4. Supervised vs. unsupervised approaches.

4.4. SLED limitations

Four main limitations of SLED warrant consideration. (1) The current model assumes the mGRE signal is a mixture of three exponential decaying components. This assumption limits SLED's ability to directly model slice profile effects that may impact 2D scans with larger excitation flip angles, as well as B_0 inhomogeneity which may produce non-exponential decays. However, as it pertains to B_0 inhomogeneity, careful iterative B_0 shimming over the whole mouse brain volume was applied prior to every mouse imaging experiment included in our study. The assumption of purely exponential decay also limits SLED's ability to analyze MWI data collected using multi-echo spin echo sequences that can be impacted by stimulated echoes. This limitation could be addressed, in the future, through the use of the extended phase graph (EPG) algorithm (Hennig, 1991; Prasloski et al., 2012). In this regard, a future extension of our study would be to develop a mathematically differentiable EPG model that could be implemented with SLED. The requirement of mathematical differentiability is necessary for current neural network training procedures (Hecht-nielsen, 1992). Additionally, diffusion and water exchange between pools during the imaging process has been omitted from the current physics model used in the decoder. Adding these biophysical effects to the SLED model is a topic for future study. (2) The global regularization property of SLED may affect the estimation accuracy of MWF for boundary values. As shown in the phantom test (Fig. 2), the MWF lower boundary (MWF = 0) was overestimated and the MWF upper boundary (MWF = 0.5) was underestimated. This phenomenon may be attributed to our chosen minimization of the SLED loss function. Specifically, the loss function was minimized in a global manner. Such a minimization approach can result in local features being over-smoothed. One potential solution for this is to incorporate the current SLED setup with convolutional neural networks (CNN) (Yamashita et al., 2018) to preserve more local features. We consider the application of CNNs for SLED an area of future research. (3) Separating the intra- and extra-cellular water pools may provide valuable information for studying various brain disorders. However, here, we did not attempt to separate such compartments due to our chosen experimental parameters and chosen magnitude MRI data analysis approach. Further discussion of our chosen pool T_2^* range estimates is given in the Supplementary Materials: 1. Justification for T_2^* constraints. Nonetheless, advanced diffusion techniques such as neurite orientation dispersion and density imaging (NODDI) (Zhang et al., 2012) could potentially be combined with SLED to attempt to distinguish the intra- and extra-cellular volume fractions and properties of the associated water pools. In this regard, combining data acquired using different MRI sequences could also be a future direction to improve the capability of SLED for more precise discrimination of the properties of intra/extra-cellular water compartments. (4) Compared to other machine learning approaches which employ synthetic decay signals to train a network then apply the trained network to experimental data, SLED trains a separate network for each subject. This leads to a somewhat higher computational cost. Although the processing time for SLED is less than 3 min, this processing time is longer than that required for application of supervised methods with trained networks. To circumvent this, high performance CPU or GPU computing could be applied in potential future applications of SLED.

5. Conclusion

This study introduced an unsupervised learning approach, SLED, which provides an accurate and practical alternative to NLLS approaches for analyzing gradient echo-based MWI data. We evaluated the performance of SLED using an *in-silico* phantom and ultra-high resolution, *ex-vivo* mouse brain imaging data. Compared to NLLS, SLED produced improved MWF maps in terms of both stability and accuracy. SLED also removed the necessity of manually selecting initialization parameters for NLLS data fitting. Advantageously, SLED is unsupervised and is trained exclusively on each dataset without the need for a comprehensive li-

brary of simulated signals. This is in contrast to the supervised learning methods currently employed for MWI data analysis. We also believe the SLED framework could be easily transformed for the data analysis of other quantitative MRI techniques.

Code availability statement

To support reproducible research, the code associated with the SLED method is made freely available at the following URL - <https://github.com/hanwencat/SLED>.

Data availability

Data are accessible after submission of a formal project outline when in keeping with McGill University's data sharing policy.

Credit authorship contribution statement

Hanwen Liu: Conceptualization, Methodology, Formal analysis, Visualization, Investigation, Resources, Software, Validation, Writing – original draft, Writing – review & editing. **Vladimir Grouza:** Methodology, Resources, Software, Writing – review & editing. **Marius Tuznik:** Methodology, Data curation, Resources. **Hooman Bagheri:** Methodology, Funding acquisition, Writing – review & editing. **Alan Peterson:** Methodology, Funding acquisition, Writing – review & editing. **David A. Rudko:** Conceptualization, Methodology, Formal analysis, Visualization, Funding acquisition, Investigation, Project administration, Resources, Software, Supervision, Writing – original draft, Writing – review & editing.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.neuroimage.2022.119717](https://doi.org/10.1016/j.neuroimage.2022.119717).

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